

Program 7

Objective: Deploy an Nginx application on Kubernetes using Minikube and access it through a NodePort service.

Project Structure

```
Pr7/  
├──  
└── nginx-deployment.yaml
```

Steps Involved

Step 1: Start Minikube

```
1rv24mc078_prateek_kumar@arch minikube start
```

Step 2: Create Deployment File

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          ports:
```

```
- containerPort: 80
```

Step 3: Apply the Deployment

```
1rv24mc078_prateek_kumar@arch kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 4: Check Deployment and Pod Status

```
1rv24mc078_prateek_kumar@arch kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	30s

```
1rv24mc078_prateek_kumar@arch kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f96d4446b-a1b2c	1/1	Running	0	25s
nginx-deployment-6f96d4446b-d3e4f	1/1	Running	0	25s

Step 5: Expose Deployment as a Service

```
1rv24mc078_prateek_kumar@arch kubectl expose deployment nginx-deployment
--type=NodePort --port=80
service/nginx-deployment exposed
```

Step 6: Check Service

```
1rv24mc078_prateek_kumar@arch kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP
nginx-deployment	NodePort	10.108.45.210	<none>	80:31234/TCP

Step 7: Access the Web Application

```
1rv24mc078_prateek_kumar@arch minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31234

Opening service default/nginx-deployment in default browser...

Step 8: Clean-up Process

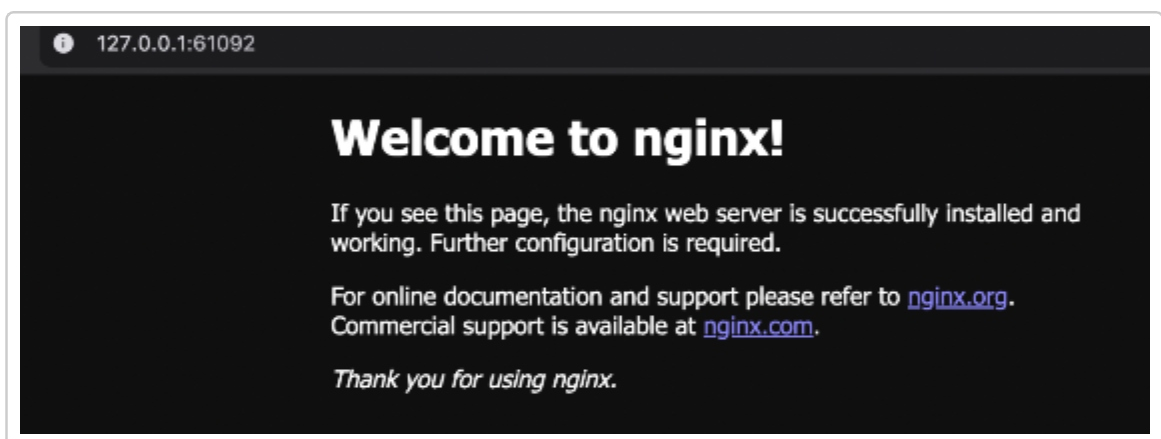
```
1rv24mc078_prateek_kumar@arch kubectl delete pods --all
pod "nginx-deployment-6f96d4446b-a1b2c" deleted
pod "nginx-deployment-6f96d4446b-d3e4f" deleted

1rv24mc078_prateek_kumar@arch kubectl delete svc nginx-deployment
service "nginx-deployment" deleted

1rv24mc078_prateek_kumar@arch kubectl delete deployments nginx-deployment
deployment.apps "nginx-deployment" deleted
```

Output

Nginx Application Running on Kubernetes:



Result

The Nginx application was successfully deployed on Kubernetes using Minikube. The deployment was exposed as a NodePort service and accessed through a web browser.